

Trajectory Reconstruction Based on IMU Data - report

So far, we have relied on an algorithm combining data from 2 IMUs, which:

- used rotation matrices to determine accelerations in the global coordinate system
- performed noise filtering using Butterworth filters / alpha filters

Observed and potential problems:

- The current rotation method was based on Euler angles and rotation matrices, which could lead to Gimbal Lock - two axes become collinear. Consequently, this results in large jumps even with small changes in measurements. Potential solution: use quaternions instead of rotation matrices and Euler angles.
- Accumulation of errors resulting from double integration (acceleration -> velocity -> position) seems inevitable if we rely solely on IMU data.
- So far, thresholds for when the robot is considered stationary have been manually set based on observations, which could lead to ignoring useful data.
- Using a Kalman filter can yield better results but requires knowledge of the object model, which seems difficult to obtain, if at all possible, since the robot will have many walking modes.
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Madgwick and Mahony filters allow reconstructing orientation based on IMU data, but unfortunately, they are unlikely to help much with position. However, they may be useful for determining whether the robot is walking on a flat surface or not:

<http://www.cs.ndsu.nodak.edu/~siludwig/Publish/papers/SPIE20181.pdf>

http://wiki.ros.org/imu_filter_madgwick

Factor graph optimization may prove useful and requires testing.